Multielectron Atoms

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Multi-electron Atoms

While exact solutions of the wave equation are not possible, approximate solution led to the following:

1. A single potential $V(r)$ may be associated with each electron. Thus we may write

$$\psi_{n,l,m_l,m_s} = R_{nl}(r)\Theta_{lm_l}(\theta)\Phi_{m_l}(\phi)\sigma_{m_s}$$

Thus each electron may be described in terms of the same set of quantum numbers used for the single electron hydrogen atom (n, l, j, m_j if spin-orbit important). The restrictions on the quantum numbers are the same.

Multi-electron Atoms

2. If the R_{nl} were computed and plotted versus r, we would find that the curves corresponding to a common value of n would exhibit a maximum at nearly the same value of r.

We thus refer to electrons with the same n as occupying the same **shell**.

Multi-electron Atoms

3. Spin-orbit coupling cannot be ignored and the energy is a function of both n and l

We refer to each l associated with a given n as a **subshell**

Shells and Subshells

- Shell Notation

Shell:	K	L	M	N	O	P…
n =	1	2	3	4	5	6…

- Subshell Notation

Subshell:	s	p	d	f	g	h…
l =	0	1	2	3	4	5…

sharp, *principal*, *diffuse*, and *fundamental*.

- To simultaneously specify shell and subshell give number and letter

3*p*, 4*s*, etc.

Building the Ground Level

- The “Ground Level” is the atom in its minimum energy configuration
- We build the atom up by noting electrons must occupy the lowest energy available subshells.
- The maximum number of electrons per subshell is determined by using the exclusion principle

The energy ordering of subshells is determined experimentally

Atomic Electronic Configurations

Atomic Number	Symbol	Configuration	Atomic Number	Symbol	Configuration
1	H	$1s^1$	11	Na	$[Ne]3s^1$
2	He	$1s^2$	12	Mg	$[Ne]3s^2$
3	Li	$[He]2s^1$	13	Al	$[Ne]3s^23p^1$
4	Be	$[He]2s^2$	Si	14	$[Ne]3s^23p^2$
5	B	$[He]2s^22p^1$	15	P	$[Ne]3s^23p^3$
6	C	$[He]2s^22p2$	16	S	$[Ne]3s^23p^4$
7	N	$[He]2s^22p3$	17	Cl	$[Ne]3s^23p^5$
8	O	$[He]2s^22p4$	18	Ar	$[Ne]3s^23p^6$
9	F	$[He]2s^22p5$	19	K	$[Ar]4s^1$
10	Ne	$[He]2s^22p6$	20	Ca	$[Ar]4s^2$

Ionization Potentials

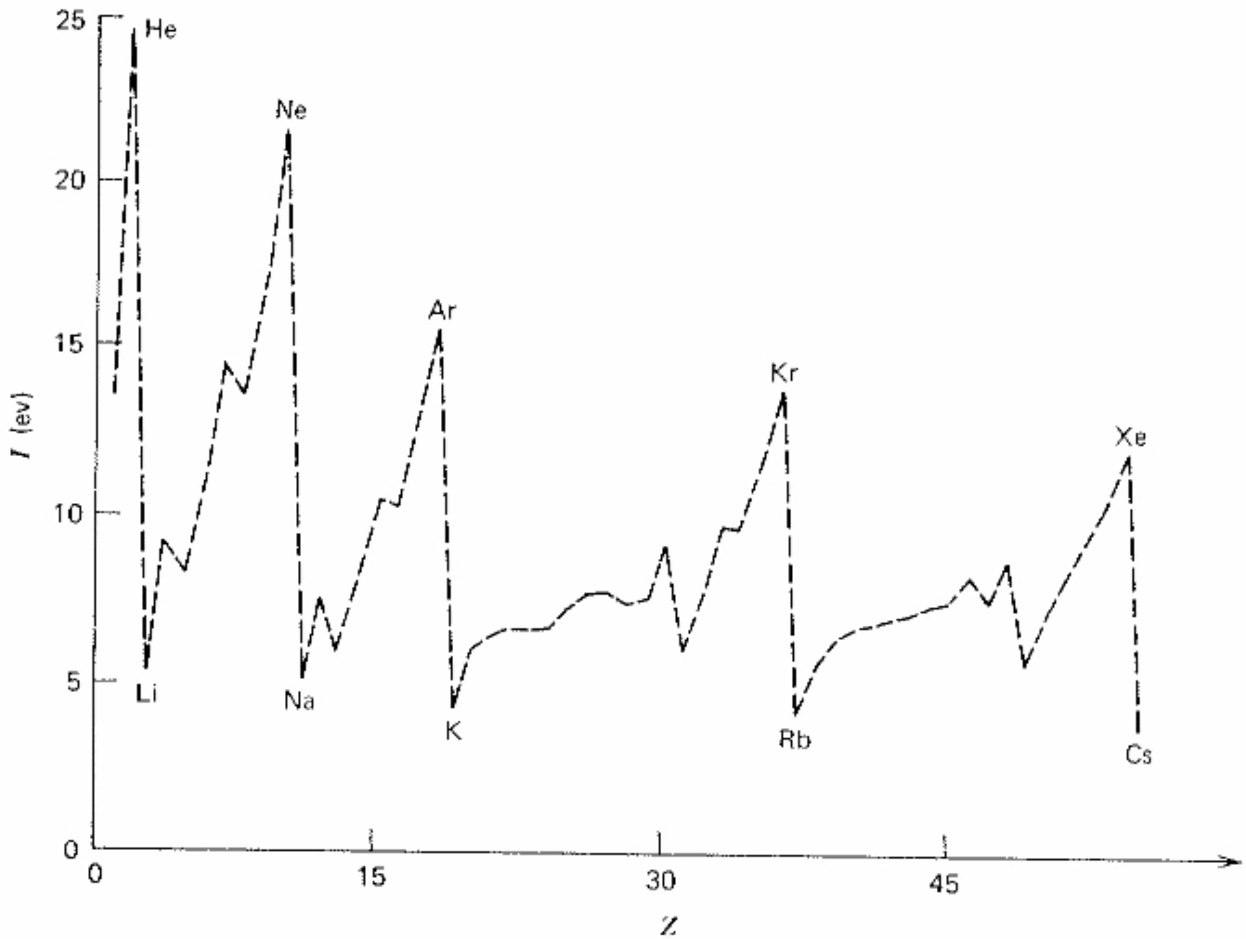

Fig. 6.5 Ionization potential as a function of atomic number for Z up to 56.

Periodic Table of the Elements 2006

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1 H 1.01																	2 He 4.00
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 15.99	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.41	31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57 La 138.91	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.2	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (270)	109 Mt (268)	110 Ds (281)	111 Rg (272)							

See "It's Elemental: The Periodic Table"
http://pubs.acs.org/cen/80th/elements.html

58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm (145)	62 Sm 150.36	63 Eu 151.97	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
90 Th 232.04	91 Pa 231.04	92 U 238.03	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Molecular Research Institute

Spectroscopic Term Classification

- We classify the electronic energy levels of an atom, referred to as **term**, by a **term symbol** of the form

$$^{2S+1}L_{J=L+S}$$

Letter:	S	P	D	F	G	H…
L =	0	1	2	3	4	5…

Multiplicity

- The multiplicity is the number of different values of J which are possible for given values of L and S. Smaller of $2S+1$ or $2L+1$
- For a given configuration and values of L and S, the different J's may have slightly different energies. For this reason we call such states **multiplets**.
  - $^4S_{3/2}$ singlet (Because L = 0)
  - $^3P_0$, $^3P_1$, $^3P_2$ triplet
  - $^4D_{1/2}$, $^4D_{3/2}$, $^4D_{3/2}$, $^4D_{7/2}$ quartet

Degeneracy

- Multiplicity should not be confused with degeneracy which here arises only from quantization of the total angular momentum along some axis.

$$J_z = m_J \hbar, \qquad m_J = -J, -J+1, \ldots, J-1, J$$

- In the absence of an external field this has no effect on energies. Thus

$$g_J = 2J + 1, \qquad g_{multiplet} = \sum_i g_{J_i}$$

Table 6.4 **The Lower Electronic Energy Levels for Selected Atoms (From C. E. Moore, *Atomic Energy Levels*, NBS Circular 467, 1949)**

	Configuration	Classification	Energy (cm^{-1})	Degeneracy
H:	$1s$	$^2S_{1/2}$	0	2
	$2p$	$^2P_{1/2,3/2}$	82,259	6
	$2s$	$^2S_{1/2}$	82,259	2
	$3p$	$^2P_{1/2,3/2}$	97,492	6
	$3s$	$^2S_{1/2}$	97,492	2
	$3d$	$^2D_{3/2,5/2}$	97,492	10
	$4p$	$^2P_{1/2,3/2}$	102,824	6
	$4s$	$^2S_{1/2}$	102,824	2
	$4d$	$^2D_{3/2,5/2}$	102,824	10
	$4f$	$^2F_{5/2,7/2}$	102,824	14
He:	$1s^2$	$^1S_0$	0	1
	$1s2s$	$^3S_1$	159,850	3
	$1s2s$	$^1S_0$	166,272	1
	$1s2p$	$^3P_{2,1,0}$	169,081	9
	$1s2p$	$^1P_1$	171,129	3
	$1s3s$	$^3S_1$	183,231	3
N:	$2s^22p^3$	$^4S_{3/2}$	0	4
	$2s^22p^3$	$^2D_{3/2,5/2}$	19,227	10
	$2s^22p^3$	$^2P_{1/2,3/2}$	28,840	6
	$2s^22p^23s$	$^4P_{1/2,3/2,5/2}$	83,320	12
	$2s^22p^23s$	$^2P_{1/2,3/2}$	86,180	6
	$2s2p^4$	$^4P_{1/2,3/2,5/2}$	88,150	12
	$2s^22p^23p$	$^2S_{1/2}$	93,582	2
	$2s^22p^23p$	$^4D_{1/2,3/2,5/2,7/2}$	94,820	20

$L = 3$

$S = 1/2$

$J = 3/2, 5/2$

$g_{3/2} = 4$

$g_{5/2} = 6$

$g_{tot} = 10$